TxBench-PP: Verifiable Evaluation of AI Agents on Preclinical Pharmacology

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ABSTRACT

We introduce TxBench-PP, a verifiable benchmark for evaluating AI agents on small-molecule preclinical pharmacology. TxBench-PP is the first focused slice of a broader TherapeuticsBench effort organized across drug-discovery stages and therapeutic modalities. It targets recurring cluster-style decisions: whether agents can recover decision-relevant conclusions from supplied assay evidence rather than recalling drug labels, ranking by a single metric, or applying generic checklists. The current task inventory is organized around five evidence sources: decryptM dose-resolved phosphoproteomics, decryptE expression proteomics, Kinobeads chemoproteomics target engagement, TG-GATEs/DrugMatrix in vivo toxicogenomics, and a curated ADMET/DMPK/PK panel. Together these datasets cover a small-molecule preclinical pharmacology spine: mechanism of action and pharmacodynamics, compound-target engagement, ADMET and PK, safety interpretation, and partial PK/PD translation. Agents receive realistic workflow snapshots, inspect files in a coding environment, and return structured answers graded deterministically. [**TODO:** Replace this final paragraph with model-run results: number of evaluations, model-harness pairs, trajectories, top pass rates, and the main failure pattern.]

TxBench-PP in TherapeuticsBench

TxBench-PP is the current small-molecule preclinical pharmacology slice of a broader TherapeuticsBench roadmap. The full benchmark family is organized by drug-discovery stage and therapeutic modality; this paper focuses on repeated, verifiable cluster decisions in small-molecule programs. These decisions sit between initial discovery and clinical development: whether a molecule engages the intended biology, whether the evidence supports its mechanism, whether exposure can cover the relevant potency range, and whether safety signals block advancement.

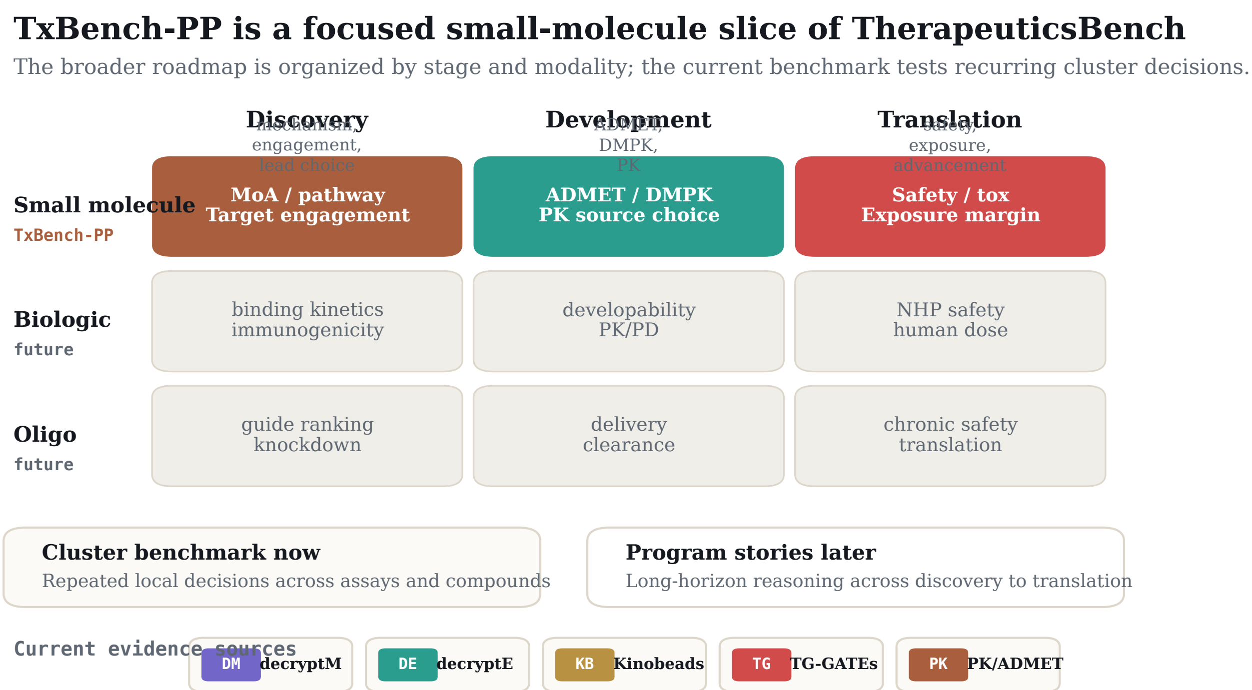

Figure 1: **TxBench-PP as a TherapeuticsBench slice.** TherapeuticsBench is intended to span discovery, development, and translation across small molecules, biologics, and oligonucleotides. TxBench-PP covers the small-molecule preclinical pharmacology slice as a cluster benchmark: repeated local decisions across assay families and compounds. Program-story benchmarks that follow one therapeutic program across discovery, development, and translation are reserved for future work.

Introduction

In preclinical drug discovery, experiments are slow and expensive, but many program-limiting decisions are interpretive: which signal is real, which compound should advance, and which liability should block a program. Preclinical pharmacology converts assay evidence into these decisions. A compound can have an attractive potency value and still be a poor candidate because the apparent mechanism is an artifact, the engaged protein is a complex contaminant, the safety signal is protocol-specific, or total exposure does not translate into unbound target coverage.

AI agents can inspect files, write analysis scripts, and invoke scientific software, but their conclusions often remain vulnerable to plausible shortcuts. In drug discovery, those shortcuts have familiar forms: importing the known mechanism of a blinded compound, trusting the largest apparent effect without quality control, ranking by potency or total exposure alone, or applying a textbook safety threshold outside the protocol where it was defined.

We introduce TxBench-PP, a benchmark for small-molecule preclinical pharmacology decisions. It should be read as a focused cluster benchmark within the broader TherapeuticsBench roadmap rather than as a complete test of drug discovery. Each evaluation supplies the agent with realistic analysis artifacts from one stage of the preclinical pharmacology evidence chain and asks for a structured, deterministically gradable conclusion. The benchmark tests whether an agent can recover the result supported by the provided evidence, not whether it can recall a drug class, run a default workflow, or optimize for a single number.

The current benchmark centers on five data sources: decryptM, decryptE, Kinobeads, TG-GATEs/DrugMatrix, and curated PK/ADMET fixtures. These datasets cover mechanism and pharmacodynamic readouts, compound-target engagement, in vivo toxicology, source reconciliation, and exposure-aware prioritization. [**TODO:** Add final run summary once result aggregation is complete.]

Benchmark Construction

To construct TxBench-PP, we decompose preclinical pharmacology workflows into short, gradeable scientific decisions. Examples include identifying a pathway-coherent mechanism from dose-resolved phosphoproteomics, separating true chemoproteomic target engagement from complex contaminants, reconciling conflicting clearance measurements, and integrating blood chemistry, pathology, and survival into an in vivo safety call.

This makes TxBench-PP a cluster-style benchmark: the same kind of decision is tested repeatedly across assay contexts, compounds, and evidence sources. Future TherapeuticsBench story evaluations can test the complementary skill of following one therapeutic program across discovery, development, and translation. TxBench-PP instead asks whether agents can make the local scientific decisions that those longer program stories depend on.

Each task includes a snapshot of assay outputs immediately before a target decision, metadata needed to interpret the files, a high-level task description, and a deterministic grader. Tasks specify what result should be recovered rather than prescribing the exact analysis method. Procedural details are included only when they are part of the scientific decision, such as whether free or total exposure should be compared to a pharmacologic threshold.

The benchmark follows three design criteria. First, tasks must be verifiable: answers are checked by structured label matching, numerical tolerances, rank-order checks, or all-of field comparisons. Second, tasks should be decision-relevant: the answer should correspond to a real go/no-go or prioritization judgment in preclinical pharmacology. Third, tasks should resist shortcuts: agents should need to inspect the supplied evidence, account for quality issues, and avoid relying on memorized drug labels or generic checklists.

We separately track decision-blocking gaps: issues that should change the program decision if recognized, such as a tox-species mismatch, insufficient free-drug coverage, a hook effect, a contaminated chemoproteomic hit, or survivor bias in pathology sampling. These labels are intended to support diagnostic analysis after endpoint grading.

Evaluation Inventory

The current suite is organized by assay evidence rather than by a generic drug discovery checklist. decryptM and decryptE map to discovery-stage mechanism, pathway, and pharmacodynamic reasoning from dose-resolved proteomic readouts. Kinobeads sits at the discovery/development bridge by testing direct target-engagement reasoning, including contaminant rejection and cluster-boundary selection. PK/ADMET tasks map to development-stage exposure, clearance, protein-binding, permeability, and source-reconciliation decisions. TG-GATEs and DrugMatrix map to translation-facing safety interpretation across chemistry, pathology, survival, and protocol context.

The inventory intentionally leaves major parts of TherapeuticsBench outside the current scope. Target genetics, disease-model characterization, broad pharmacogenomics, primary screening QC, biologics developability, antibody-drug conjugates, degraders, oligonucleotides, and cell or gene therapies are natural future extensions, but are not required to define the small-molecule preclinical pharmacology slice tested here.

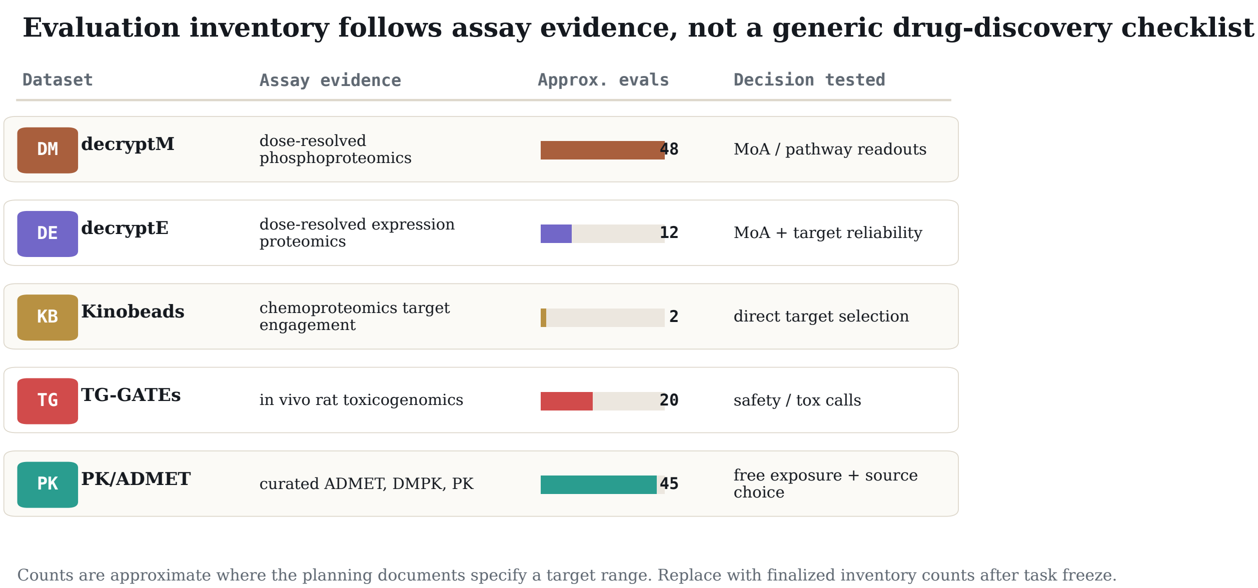

Figure 2: Evaluation inventory by evidence source. Current TxBench-PP tasks are drawn from five small-molecule preclinical pharmacology evidence sources. Approximate evaluation counts come from the planning documents and should be replaced with frozen counts before submission.

Evidence source	Evals	Assay evidence	Decision themes
decryptM	48	dose-resolved phosphoproteomics	Mechanism-of-action inference, pathway-coherent signalling, artifact rejection
decryptE	12	dose-resolved expression proteomics	Expression-proteomic pathway response, reliable target signal selection, single-feature artifact avoidance
Kinobeads	2	chemoproteomics target engagement	Direct target selection, contaminant removal, engaged-cluster boundary detection
TG-GATEs / DrugMatrix	∼20	in vivo rat toxicogenomics and safety	Integrated toxicity calls, pathology and chemistry reconciliation, survivor bias, species-specific artifacts
PK / ADMET panel	∼45	ADMET, DMPK, PK, literature measurements	Clearance source reconciliation, free-drug exposure, safety margins, potency-adjusted prioritization

Table 1: Draft TxBench-PP inventory. The current benchmark should be described as a small-molecule preclinical pharmacology suite. Counts marked with ∼ should be replaced when the final evaluation manifest is frozen.

Results

Agents do not yet reliably recover preclinical pharmacology decisions

[**TODO:** Insert topline result paragraph. Recommended structure: state the number of model-harness pairs, total trajectories, pass-rate definition, top four systems with percentage first and counts/CIs in parentheses, then explain whether the leading systems form a broad tier or a stable ordering.]

[**TODO:** Insert a horizontal model leaderboard figure after result tables exist. Use the EpiBench bar-plot orientation and report pass rate as percentage with fractions and confidence intervals in the table.]

Failures reflect program-relevant judgment gaps

TxBench-PP is designed so that many failures should be interpretable as drug-program judgment failures rather than arbitrary execution failures. The expected failure families are: literature priors over assay evidence, assay-context calibration failures, missing multi-property integration, and rigid thresholds that ignore margin or protocol context.

[**TODO:** Replace this boilerplate with trajectory-derived counts after manual review. Report the number of reviewed evaluations and make clear that these labels are diagnostic rather than exhaustive.]

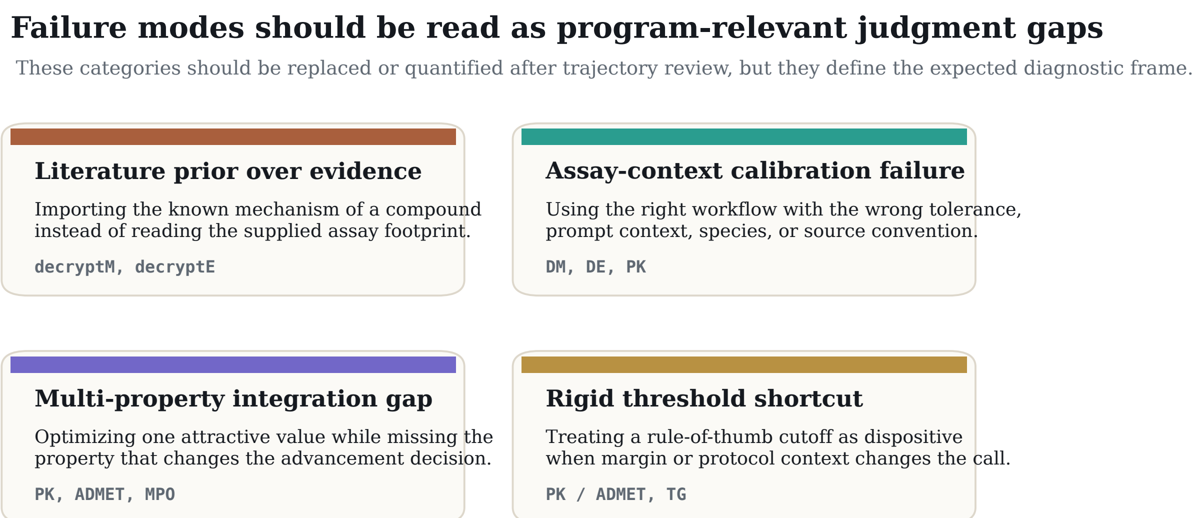

Figure 3: Expected failure taxonomy. TxBench-PP tasks are designed to distinguish careful preclinical pharmacology reasoning from program-relevant judgment gaps: literature priors overriding assay evidence, assay-context calibration failures, missing multi-property integration, and rigid thresholds. Replace this design taxonomy with empirically tagged failure counts after trajectory review.

Case examples span the compound-centered preclinical spine

The benchmark examples should make the task format concrete. In decryptM, a blinded kinase inhibitor can carry a familiar class label while the dose-resolved phosphorylation footprint supports a different kinase-family mechanism. In Kinobeads, the strongest apparent protein signal may be a complex contaminant rather than a direct binder. In TG-GATEs, a blood-chemistry threshold may be less informative than pathology and survival under the study protocol. In PK tasks, total exposure or potency alone can select a compound that lacks adequate unbound coverage over the dosing interval.

[TODO: Replace these examples with exact task names, final answer surfaces, and short descriptions after the public task manifest is frozen.]

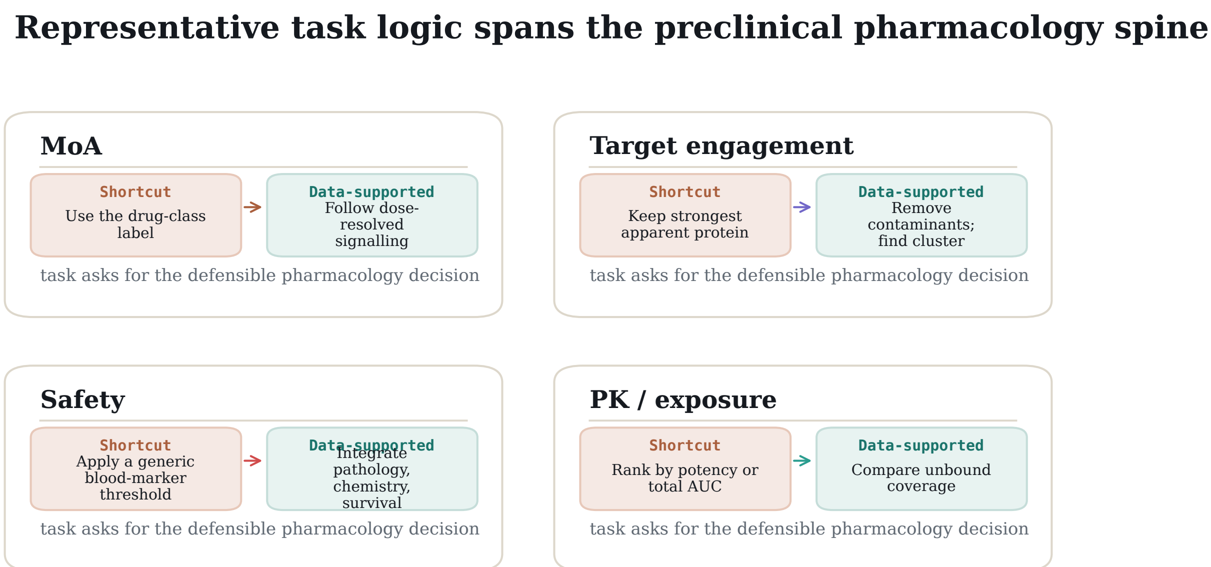

Figure 4: Representative task logic. Across mechanism, target engagement, safety, and exposure decisions, the correct answer is the one supported by the supplied assay evidence, not the familiar shortcut.

Discussion

TxBench-PP measures a narrow but important part of drug discovery: whether an agent can turn preclinical pharmacology evidence into a defensible local decision. This scope is intentionally smaller than end-to-end therapeutic discovery. It is a small-molecule cluster benchmark within the broader TherapeuticsBench roadmap, not a claim to cover every stage or modality. It does not yet test target genetics, full disease-model selection, primary screening operations, broad clinical translation, or non-small-molecule modalities. The narrower scope makes the benchmark verifiable while still covering decisions that directly affect compound advancement.

The central hypothesis is that current agents will often operate the tools of preclinical analysis but fail at the scientific judgment layer. If a model selects a mechanism from a drug label, trusts a contaminated target-engagement signal, ignores survivor bias, or ranks by total exposure rather than unbound coverage, it may produce a fluent and computationally elaborate answer that is not the pharmacology decision supported by the data.

[**TODO:** After results are available, add a paragraph comparing endpoint pass rates, partial-credit or field-level scores if available, and trajectory review. Emphasize what capability is missing and which tasks show meaningful progress.]

Future versions can extend the benchmark upstream to target rationale and primary screening, sideways to pharmacogenomics and disease-model characterization, and across modalities to antibodies, ADCs, degraders, oligonucleotides, cell therapies, and gene therapies. A complementary long-horizon TherapeuticsBench track can follow program stories across discovery, development, and translation. For the current paper, the clean claim should remain: TxBench-PP is a verifiable cluster benchmark for small-molecule preclinical pharmacology decisions.

Methods

Task construction

Candidate evaluations are derived from realistic preclinical pharmacology workflow states. A task is retained when the target conclusion can be recovered from the supplied files, expressed through a constrained final answer, and graded deterministically. Tasks should be revised or removed when the prompt over-specifies the method, when multiple reasonable analyses produce incompatible decisions, or when the grader cannot distinguish the supported pharmacology conclusion from a plausible but unsupported answer.

Agent runs and aggregation

[**TODO:** Insert model-harness list, number of attempts per evaluation, execution environment, parsing rules, and how invalid or missing outputs are counted.]

Statistical analysis

[**TODO:** Insert primary metric and confidence interval convention. If following SpatialBench/EpiBench, report evaluation-weighted endpoint pass rate, raw pass counts for interpretability, and Student t confidence intervals over evaluation-level replicate means.]

Data availability

[**TODO:** Insert public repository, public example evaluations, result summaries, and trajectory availability language.]

References

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